

DATA ANALYTICS INTERNSHIP - InternSpark

(AICTE-Listed Internship Program)

REPORT **ON** **CUSTOMER BEHAVIOR** **ANALYSIS** **(TASK 1)**

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This report presents a comprehensive analysis of customer behavior using transactional and demographic data to derive meaningful business insights for Alfido Tech. The primary objective of this study is to understand purchasing patterns, segment customers effectively, and identify factors influencing customer churn.

The dataset includes key attributes such as customer demographics, purchase history, product categories, payment methods, and return behavior. After performing data cleaning and preprocessing, advanced analytical techniques such as RFM (Recency, Frequency, Monetary) analysis were applied to segment customers into distinct groups, including Best Customers, Loyal Customers, Recent Customers, Big Spenders, and At-Risk Customers.

The analysis reveals that a small segment of high-value customers contributes significantly to overall revenue, emphasizing the importance of customer retention strategies. Additionally, purchasing behavior varies across product categories and time periods, indicating potential seasonal trends and category-specific performance. Payment method preferences and return patterns further highlight areas where customer experience can be optimized.

Churn analysis indicates that customers with lower purchase frequency and spending are more likely to disengage. This highlights the need for proactive strategies to retain customers and improve long-term engagement.

Based on these insights, several recommendations have been proposed, including the implementation of targeted loyalty programs, personalized marketing strategies, optimization of product offerings, and enhancement of the payment and checkout experience. These data-driven strategies aim to improve customer satisfaction, reduce churn, and maximize revenue.

INTRODUCTION

This task focuses on analyzing customer transaction data to understand purchasing behavior, segment customers, and identify churn risks. The objective is to derive meaningful insights that can help improve customer engagement and business performance for Alfido Tech.

OBJECTIVES

1. Analyze customer purchase behavior.
2. Segment customers using RFM analysis.
3. Identify purchase patterns and trends.
4. Detect churn behavior and risks.
5. Provide actionable business recommendations.

DATASET DESCRIPTION

The dataset contains customer transaction and demographic details with the following key attributes:

1. Customer ID

2. Purchase Date
3. Product Category
4. Product Price
5. Quantity
6. Total Purchase Amount
7. Payment Method
8. Age, Gender
9. Returns
10. Churn

DATA CLEANING AND PREPROCESSING

The dataset was cleaned and prepared before analysis:

- Removed duplicate and unnecessary columns (e.g., duplicate age column).
- Converted Purchase Date into datetime format.
- Checked and handled missing values.
- Ensured valid values for price and quantity.
- Created structured dataset for analysis.

FEATURE ENGINEERING

New features were created to improve analysis:

1. Extracted Month and Year from Purchase Date.
2. Verified Total Purchase Amount.
3. Created customer-level aggregated data.
4. Prepared dataset for RFM segmentation.

CUSTOMER SEGMENTATION (RFM ANALYSIS)

Customers were segmented using:

- **Recency (R):** Days since last purchase
- **Frequency (F):** Number of purchases
- **Monetary (M):** Total spending

Segments Identified:

1. Best Customers
2. Loyal Customers
3. Recent Customers
4. Big Spenders
5. At Risk Customers

This segmentation helps identify high-value and low-engagement customers.

PURCHASE BEHAVIOR ANALYSIS

Key observations from purchase patterns:

- Certain product categories generate higher revenue.
- Sales show variation across months, indicating seasonality.
- Customers prefer specific payment methods.
- Repeat customers contribute significantly to revenue.

CHURN ANALYSIS

Customer churn behavior was analyzed to identify risks:

- A portion of customers have stopped purchasing.
- Customers with low frequency and low spending show higher churn.
- Spending patterns differ between churned and retained customers.
- Age groups show variation in churn behavior.

SCREENSHOTS AND VISUALIZATIONS

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

In [2]: df = pd.read_csv("ecommerce_customer_data_custom_ratios.csv")

In [3]: df.head()

Out[3]:
```

	Customer ID	Purchase Date	Product Category	Product Price	Quantity	Total Purchase Amount	Payment Method	Customer Age	Returns	Customer Name	Age	Gender	Churn
0	46251	2020-09-08 09:38:32	Electronics	12	3	740	Credit Card	37	0.0	Christine Hernandez	37	Male	0
1	46251	2022-03-05 12:56:35	Home	468	4	2739	PayPal	37	0.0	Christine Hernandez	37	Male	0
2	46251	2022-05-23 10:18:01	Home	288	2	3198	PayPal	37	0.0	Christine Hernandez	37	Male	0
3	46251	2020-11-12 13:13:29	Clothing	196	1	3509	PayPal	37	0.0	Christine Hernandez	37	Male	0
4	13593	2020-11-27 17:55:11	Home	449	1	3452	Credit Card	49	0.0	James Grant	49	Female	1

```
In [4]: df.columns

Out[4]: Index(['Customer ID', 'Purchase Date', 'Product Category', 'Product Price',
              'Quantity', 'Total Purchase Amount', 'Payment Method', 'Customer Age',
              'Returns', 'Customer Name', 'Age', 'Gender', 'Churn'],
              dtype='object')
```

```
In [5]: df = df.drop(['Customer Age'], axis=1)
df['Purchase Date'] = pd.to_datetime(df['Purchase Date'])
df.isnull().sum()
```

```
Out[5]: Customer ID      0
Purchase Date      0
Product Category    0
Product Price      0
Quantity           0
Total Purchase Amount 0
Payment Method     0
Returns           47596
Customer Name      0
Age               0
Gender            0
Churn             0
dtype: int64
```

```
In [6]: df['Total Purchase Amount'].sum()
```

```
Out[6]: 681342683
```

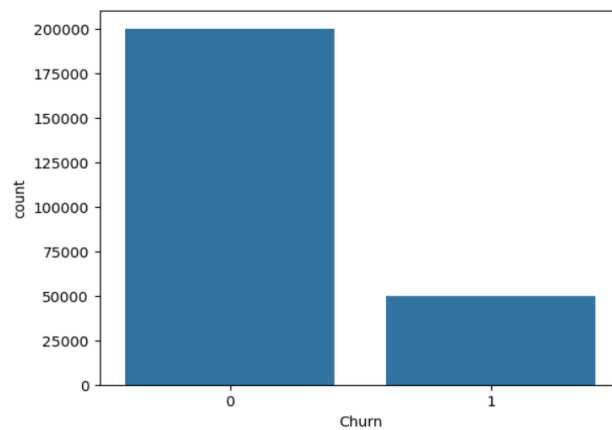
```
In [7]: df['Total Purchase Amount'].mean()
```

```
Out[7]: 2725.370732
```

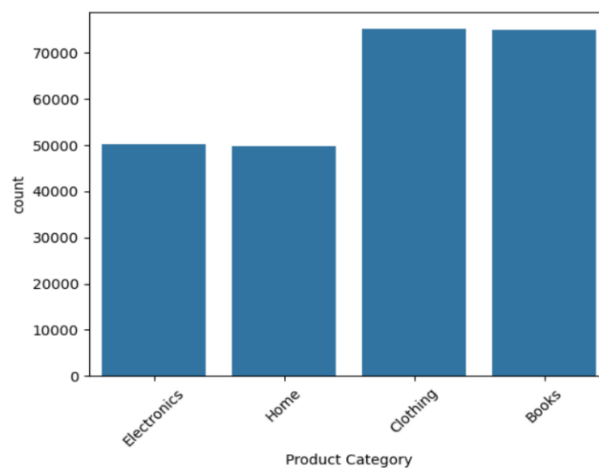
```
In [8]: df['Product Category'].value_counts()
```

```
Out[8]: Product Category
Clothing    75052
Books       74912
Electronics 50185
Home        49851
Name: count, dtype: int64
```

```
In [10]: sns.countplot(data=df, x='Churn')
plt.show()
```



```
In [9]: sns.countplot(data=df, x='Product Category')
plt.xticks(rotation=45)
plt.show()
```



```
In [11]: snapshot_date = df['Purchase Date'].max() + pd.Timedelta(days=1)
```

```
In [12]: rfm = df.groupby('Customer ID').agg({
    'Purchase Date': lambda x: (snapshot_date - x.max()).days,
    'Customer ID': 'count',
    'Total Purchase Amount': 'sum'
})
rfm.rename(columns={
    'Purchase Date': 'Recency',
    'Customer ID': 'Frequency',
    'Total Purchase Amount': 'Monetary'
}, inplace=True)
rfm.head()
```

```
Out[12]:
```

Customer ID	Recency	Frequency	Monetary
1	58	1	3491
2	299	3	7988
3	89	8	22587
4	127	4	8715
5	171	8	12524

```
In [13]: rfm['R_score'] = pd.qcut(rfm['Recency'], 4, labels=[4,3,2,1])
rfm['F_score'] = pd.qcut(rfm['Frequency'], 4, labels=[1,2,3,4])
rfm['M_score'] = pd.qcut(rfm['Monetary'], 4, labels=[1,2,3,4])
```

```
In [14]: rfm['RFM_Score'] = rfm['R_score'].astype(str) + rfm['F_score'].astype(str) + rfm['M_score'].astype(str)
```

```
In [15]: def segment(row):
    if row['RFM_Score'] == '444':
        return 'Best Customers'
    elif row['R_score'] == 4:
        return 'Recent Customers'
    elif row['F_score'] == 4:
        return 'Loyal Customers'
    elif row['M_score'] == 4:
        return 'Big Spenders'
    else:
        return 'At Risk'

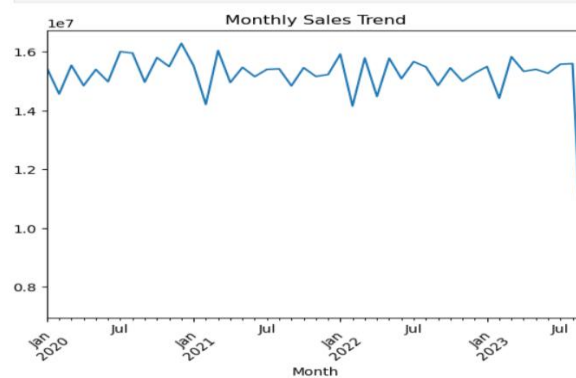
rfm['Segment'] = rfm.apply(segment, axis=1)
```

```
In [16]: rfm['Segment'].value_counts()
```

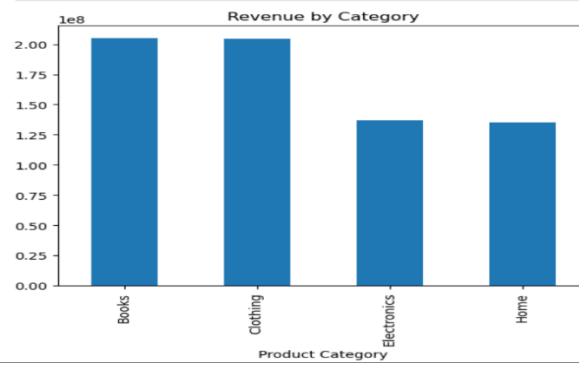
```
Out[16]: Segment
At Risk          27472
Recent Customers  8948
Loyal Customers  7343
Best Customers   3569
Big Spenders     2341
Name: count, dtype: int64
```

```
In [18]: df['Month'] = df['Purchase Date'].dt.to_period('M')
monthly_sales = df.groupby('Month')['Total Purchase Amount'].sum()

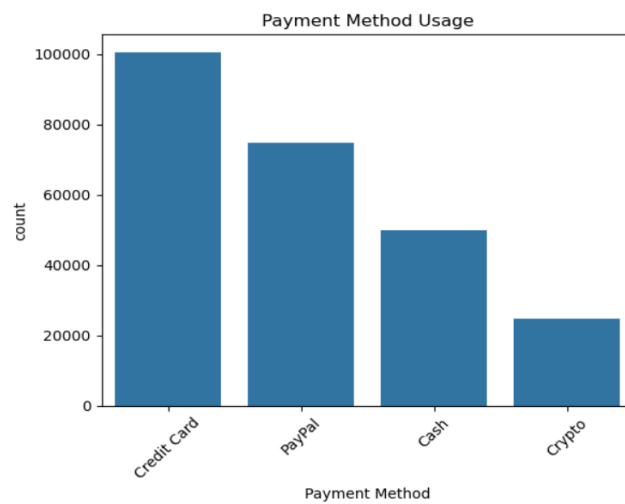
monthly_sales.plot()
plt.title("Monthly Sales Trend")
plt.xticks(rotation=45)
plt.show()
```



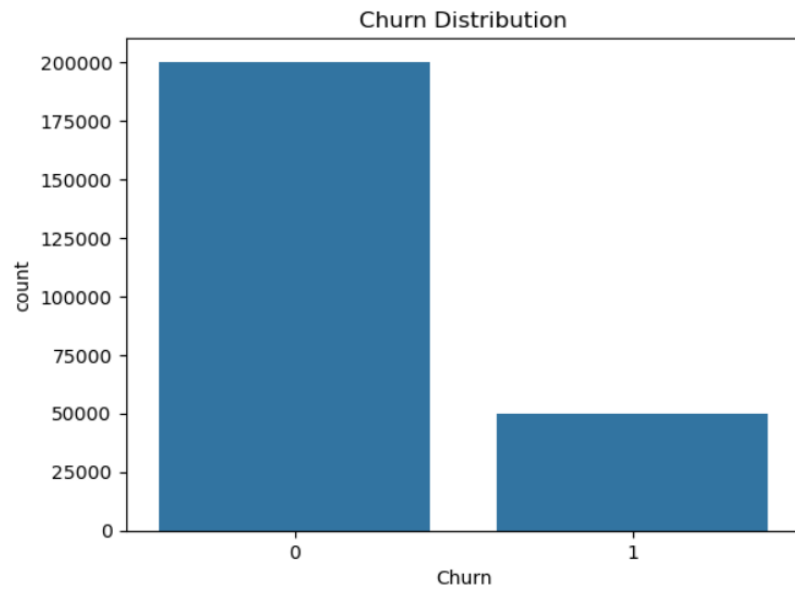
```
In [19]: category_sales = df.groupby('Product Category')['Total Purchase Amount'].sum()
category_sales.sort_values(ascending=False).plot(kind='bar')
plt.title("Revenue by Category")
plt.show()
```



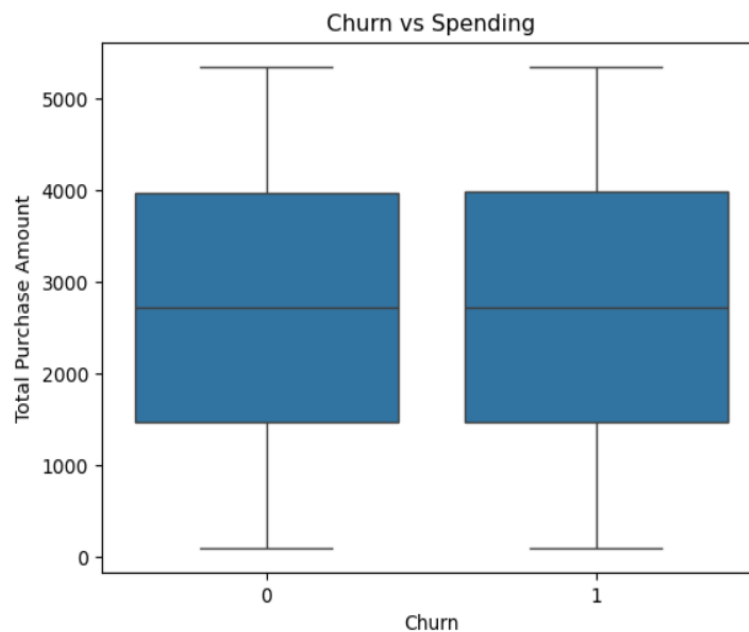
```
In [23]: sns.countplot(data=df, x='Payment Method')
plt.title("Payment Method Usage")
plt.xticks(rotation=45)
plt.show()
```



```
In [24]: sns.countplot(data=df, x='Churn')
plt.title("Churn Distribution")
plt.show()
```



```
In [26]: sns.boxplot(data=df, x='Churn', y='Total Purchase Amount')
plt.title("Churn vs Spending")
plt.show()
```



```
In [27]: churn_rate = df['Churn'].value_counts(normalize=True) * 100
churn_rate
```

```
Out[27]: Churn
0      80.0504
1      19.9496
Name: proportion, dtype: float64
```


KEY INSIGHTS

- High-value customers contribute a major share of revenue.
- A small segment of customers are highly loyal and frequent buyers.
- Customers with fewer purchases are more likely to churn.
- Product category performance varies significantly.
- Payment method trends indicate customer preferences.

RECOMMENDATIONS

1. **Implement targeted loyalty programs** : High-value customers identified as “Best” and “Loyal Customers” contribute significantly to overall revenue. Alfido Tech should introduce structured loyalty programs such as reward points, exclusive discounts, early access to new products, or premium memberships. These initiatives will strengthen customer relationships, encourage repeat purchases, and increase long-term customer lifetime value.
2. **Develop a proactive churn prevention strategy** : Customers categorized as “At Risk” show declining engagement and purchasing frequency. The company should proactively target these customers through personalized email campaigns, special discounts, and reactivation offers. Additionally, analyzing their past behavior can help identify reasons for disengagement and allow the business to take corrective actions before losing them permanently.
3. **Leverage personalization for better engagement** : Customer data such as purchase history, age, gender, and product preferences can be used to deliver personalized recommendations. By implementing recommendation systems (e.g., “Customers who bought this also bought...”), Alfido Tech can enhance user experience, increase cross-selling and upselling opportunities, and improve overall customer satisfaction.
4. **Focus on high-performing product categories** : The analysis shows that certain product categories generate a larger share of revenue. Alfido Tech should prioritize these categories by increasing inventory, optimizing pricing strategies, and running targeted marketing campaigns. At the same time, underperforming categories should be evaluated for improvement or repositioning to maximize profitability.
5. **Optimize payment experience and checkout process** : Customer preference for specific payment methods indicates the importance of a smooth and flexible checkout process. Alfido Tech should ensure seamless integration of popular payment options, reduce transaction failures, and enhance user experience during checkout. A simplified and secure payment process can significantly improve conversion rates and customer satisfaction.

CODE LINK : https://github.com/shivangikumari1409/InternSpark-Internship/blob/ea879ea2f24d16efa638d9923bf3042f80b28655/CustomerBehaviorAnalysis/Customer_Behavior_Analysis.ipynb